Rahul Viswanath

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EDUCATION

Bentonville High School — Bentonville, AR
Weighted GPA: 4.42 | Class Rank: 11 of 867 — Expected May 2027
Ignite Professional Studies - Technology Strand — Fall 2025 - Spring 2027
Concurrent Enrollment: NorthWest Arkansas Community College (NWACC), Bentonville, AR
Relevant Coursework: AP Computer Science Principles; AP Computer Science A

EXPERIENCE

Computer Infrastructure Engineering Intern — June 2026 - Present
Thorbole Simulation Technologies LLC - Crash-Safety Simulation — Bentonville, AR (Hybrid)

- Built a hybrid-search knowledge library for the company's Dropbox archive using PyMuPDF for PDF extraction, Tesseract OCR for scanned files, sentence-transformers embeddings, Qdrant vector retrieval, and Meilisearch keyword search.
- Automated crash-image analysis and descriptions with the Claude Vision API to reduce repetitive manual image review by engineers.
- Developed a Dropbox Model Context Protocol (MCP) integration that lets AI assistants directly search, read, and work with company files.
- Designed and assembled a high-performance workstation with an NVIDIA GeForce RTX 5090 (32GB GDDR7), 128GB DDR5-6400, Samsung 990 EVO Plus PCIe 4.0 NVMe storage, and Corsair HX1200i 1200W Platinum power for large-scale simulation, 4K/8K rendering, CUDA-accelerated AI, and large-model inference.

Founder and Technical Lead — September 2025 - Present
PATHOS - Independent Assistive Technology Project — Bentonville, AR

- Founded an affordable, non-diagnostic stress-awareness wearable for autistic and nonverbal people; led hardware, machine learning, responsible-use language, outreach, and national competition preparation.
- Built the new end-to-end pipeline: EDA and skin-temperature signals → noise filtering → 60-second windows → 43-feature extraction and schema checks → ensemble inference → live Streamlit stress-probability dashboard; evaluated 865 WESAD windows across 15 participants using leave-one-subject-out validation (~77% accuracy).
- Demonstrated a working prototype as an AIMFest vendor to 750+ attendees; earned 5NEWS coverage, mentorship from Alice Walton School of Medicine, and nonprofit validation through Autism Involves Me, which serves 2,000+ people.
- Expanded public reach through @autism_feed coverage that generated 13.6K+ likes, 4,389 saves, and 1,009 reposts.

Former Chief AI Officer — September 2025 - August 2026
EverGreen - Independent AI Sustainability Project — Bentonville, AR

- Built a custom YOLO vision system for a smart waste-sorting prototype, with 2.5 ms inference and 97%+ target accuracy for PET plastic and aluminum classification.
- Qualified for the international EarthX e-Capital Summit, collaborated with Startup Junkie and Bentonville stakeholders, and contributed AI architecture and model development to EverGreen's signed patent materials.

Marketing Intern and Tech Lead — January 2026 - Present
Women+ In Tech (WIT+) — Bentonville, AR

- Coordinated events and outreach, led a seven-month social media campaign, and designed AI/ML and engineering workshops for Girls+ in Tech.

LEADERSHIP AND ACTIVITIES

Reporter, SkillsUSA Club — August 2025 - Present

- Managed grants, fundraising, state conference operations, and member communications for a chapter of 65+ students.

Secretary, Tech4All Club — August 2025 - Present

- Coordinated meetings, officer communications, and guest lectures with WIT+ sponsors.

AWARDS AND HONORS

Awards: 10th Place Nationally, Community Action Project, SkillsUSA National Leadership & Skills Conference, Atlanta, GA (2026); EarthX e-Capital Summit International Qualifier (2026)
Certifications: IBM Emerging Technology Certification; IBM Cybersecurity Fundamentals

TECHNICAL SKILLS

Programming and Web: Python, C++, Java, JavaScript, React, HTML/CSS
AI, Search, and Data: LLMs, computer vision, OCR, embeddings, vector databases, sentence-transformers, Qdrant, Meilisearch, PyTorch, scikit-learn, TensorFlow, XGBoost, LightGBM, Pandas, NumPy
Automation and Systems: Model Context Protocol, Claude Vision API, Streamlit, Dropbox, CUDA, workstation integration, API integration, Git/GitHub, Linux, Jupyter
Embedded Systems: ESP32/ESP32-C3, Arduino, Raspberry Pi, I2C, BME280, EDA/GSR, CAD, PySerial
Languages: English (native), Tamil, Hindi, Kannada, French